



Foundation GEOINT Modernization (FG Mod)

Program Overview

June 2022

The overall classification of this presentation is:
UNCLASSIFIED

Classified by:
Derived from:
Declassify on:

N A T I O N A L G E O S P A T I A L **NGA** I N T E L L I G E N C E A G E N C Y

Program Objectives

- Support Foundation GEOINT Group Mission – Timely, Relevant, & Accurate...
- Use Agile development methodology
- Focus on Initial Operating Capabilities first, work towards Full Operating Capability
- Leverage COTS technology to accelerate modernization
- Establish an enterprise data management architecture in the Cloud
- Standardize production to improve quality & consistency
- Move production to UNCLASS
- Establish authoritative sources & services
- Enable Foundation GEOINT Group content for broader mission use

NGA Source Directorate – Foundation GEOINT Group



Scaled Agile Development – Mission Focused Scrum Teams

Maritime

Enterprise Data Management

Chart Automation

Workflow Management

NITE

Boundaries

Web-based Dissemination

Workflow Management

Advanced Editing Tools

Topography

Product on Demand

Cartographic Production

Workflow Management

Imagery

Ortho on-the-fly

NITE

Catalog Search

Data Analytics

Stats

Content Reports

Suitability

Human Geography

Data Integration

Places Management

Data Analysis

Requirements Management

Web-based Collection

Reporting

Prioritization

Publications

Sailing Directions

List of Lights

World Port Index

GeoNames

Names Management

Workflow Management

Web-based Dissemination

Vector Tiles

Data Integration

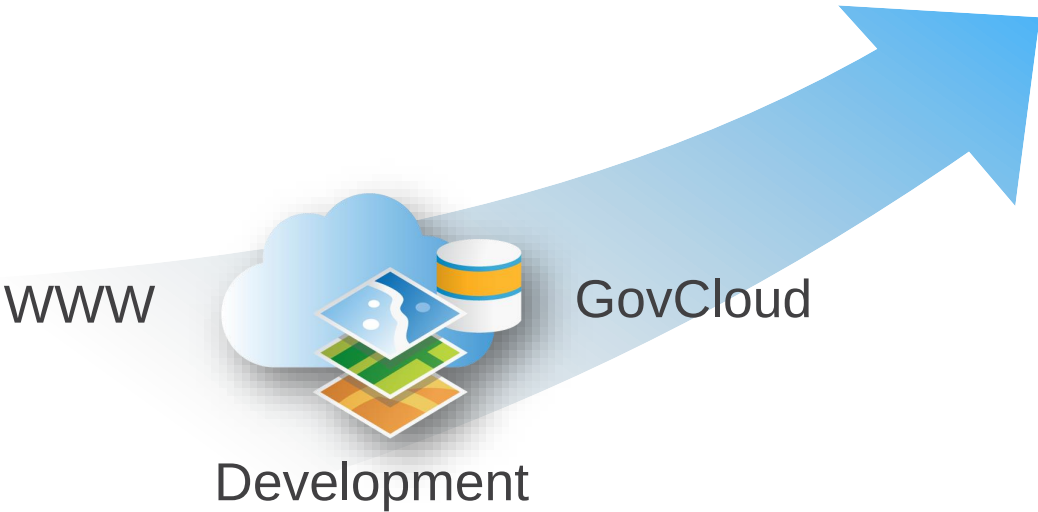
Vector Tile Basemap Production

Map Styles: Canvas Analytic, Scanned Maps, Greenmap Vector Tile, Live Light Gray Vector Tile, Live Light Vector Tile, Live Bright Vector Tile, Light Gray Vector Tile, Dark Vector Tile, Monochrome Vector Tile



Cloud Architecture – Producing and Sharing Data Across Security Networks

- Capabilities
- 14 Cloud Environments
 - System of Systems Architecture
 - PostgreSQL RDS
 - Citrix Virtual Desktops



TOP SECRET



C2S (TC)
TP1, TP2, OPS

Replicate

SECRET



SC2S (SC)
TP1, TP2, OPS

Replicate

UNCLASS



UC2S (UC)
TP1, TP2, OPS

Replicate

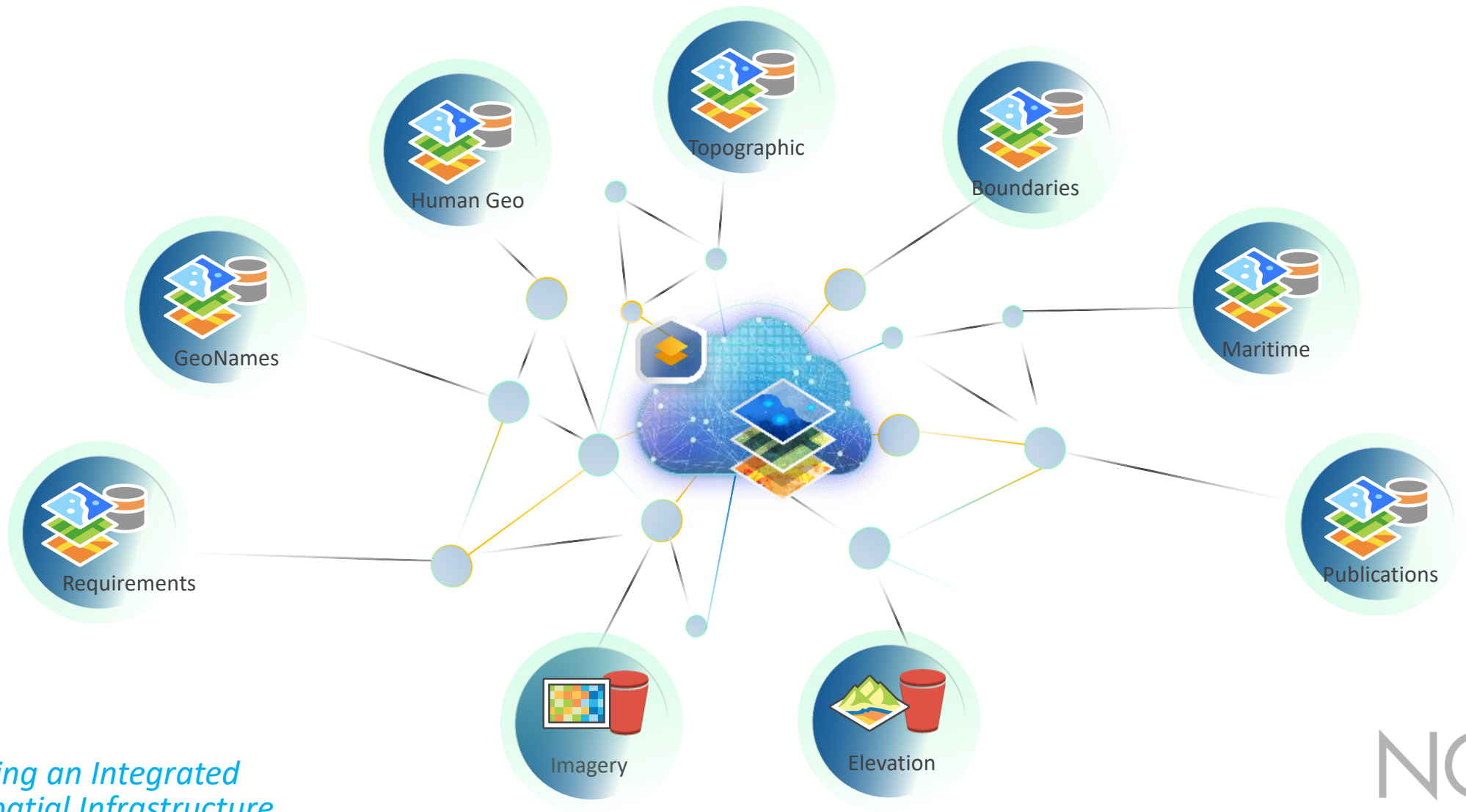
WWW



NC2S (NC)
TP1, TP2, OPS



Foundation GEOINT Enterprise GIS – Integrated, Interoperable & Extensible



*Creating an Integrated
Geospatial Infrastructure*

Foundation GEOINT Modernization - Capabilities Status

Targeted Capabilities

Maritime Safety Office (SFH)

- 1. Digital Nautical Charts (DNC)
- 2. Electronic Navigational Charts S-57 (ENC)
- 3. Standard Nautical Charts (SNC)
- 4. Tactical Ocean Data (TOD)
- 5. Non-Submarine Contacts (NSC)
- 6. Mobile Offshore Drilling Units (MODU)
- 7. Anti-Shipping Activity Message (ASAM)
- 8. List of Lights (LoL)
- 9. World Port Index (WPI)
- 10. Sailing Directions (SD)
- 11. Submarine ENC (SMENC)
- 12. Littoral Planning Charts (LPC)
- 13. Additional Military Layers (AML)
- 14. Radio Navigation Aids (RNA)
- 15. Electronic Navigational Charts S-100 (ENC)

Office of Geography (SFG)

- 16. PLACES
- 17. Boundaries and Limits
- 18. Topo Vector Data Production
- 19. Topo Cartographic Production
- 20. Human Geography
- 21. GeoNames
- 22. State of the Data
- 23. Georeferenced Map Archive

Office of Geomatics (SFN)

- 25. Elevation Services
- 26. NTM Imagery Services
- 27. Commercial Imagery Services
- 28. World Imagery Wayback
- 29. GEOINT Research Tools

NOX Office (SFP)

- 30. System for Requirements Management

Data Integration and Automation Office (SFD)

- 32. Vector Tiles

Operational Capabilities

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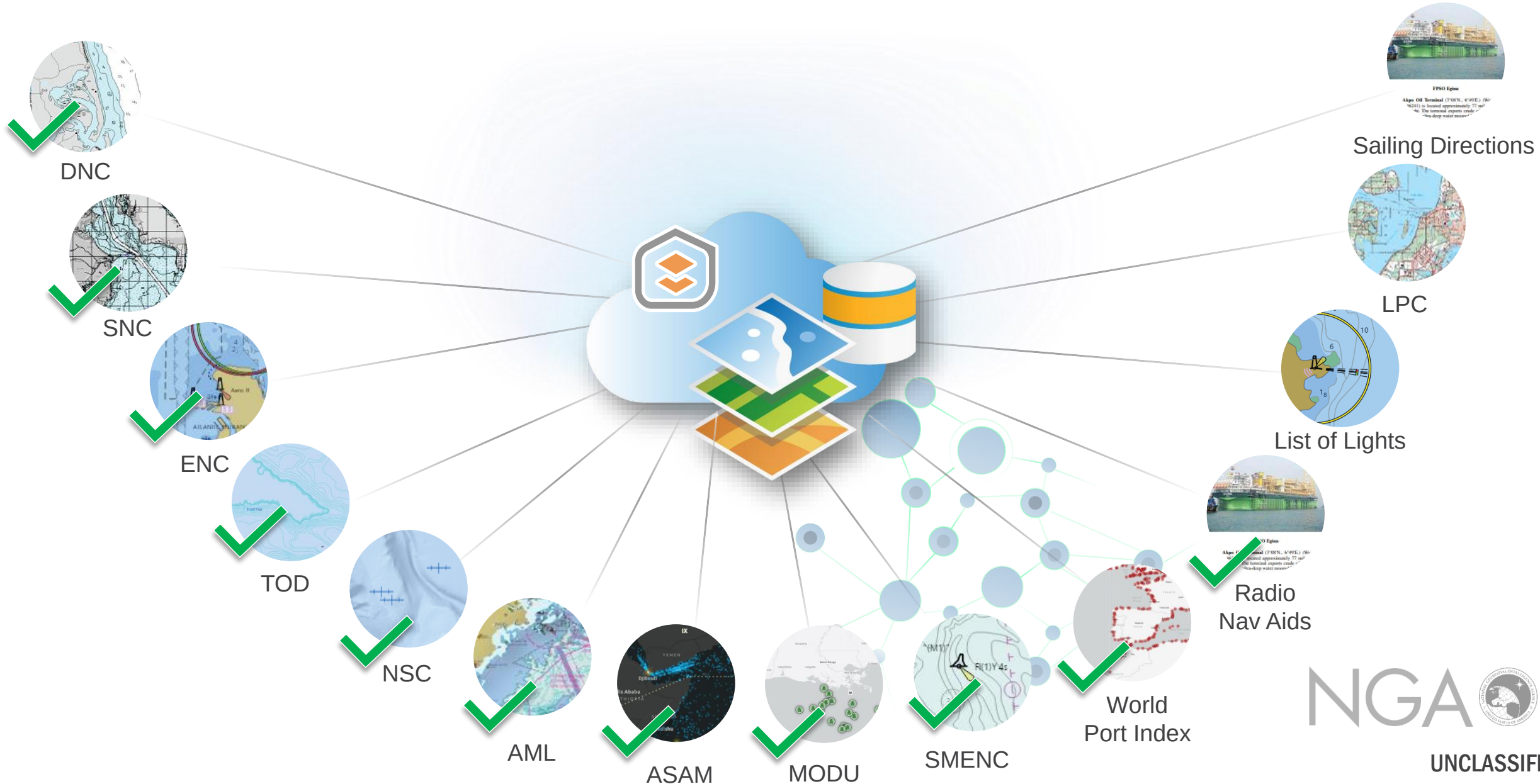
Data Integration and Automation Office (SFD)

- 32.



Maritime Safety Office

NGA Maritime Cloud – Data Centric Enterprise Hydrographic Production

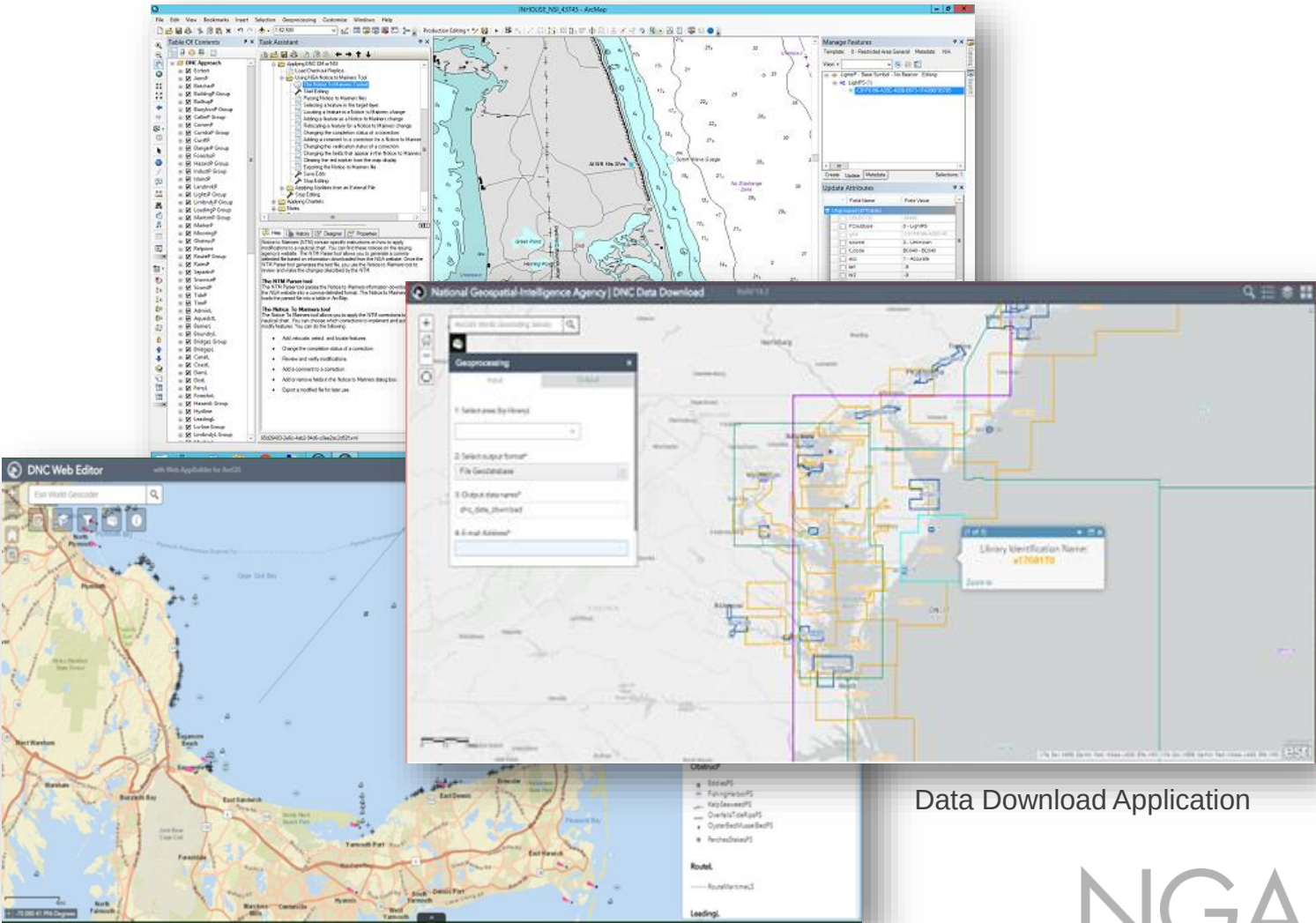


Digital Nautical Charts - Supporting Maritime Safety of Navigation

Capabilities:

- Enterprise Geodatabase
- Virtualized Production Desktop
- Structured Workflows
- Job Management
- Quality Control
- Web Editing Application
- MSPS Integration
- Dissemination Services
- Data Download Web App

Desktop Production



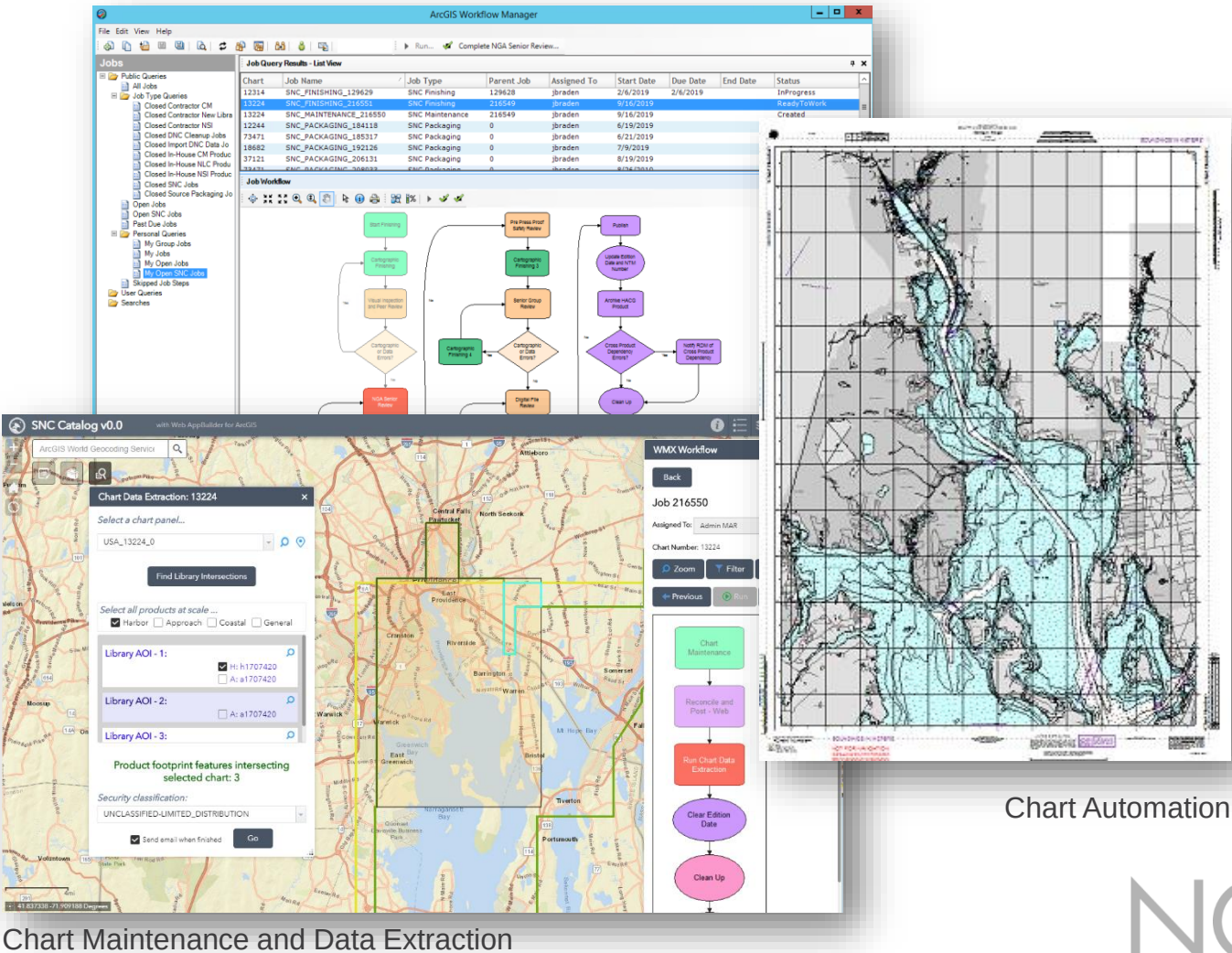
Data Download Application

Web Editing



Workflow Management and Metrics

- Virtualized Production Desktop
- Chart Automation Tools
- SNC Cartography
- Structured Workflows
- Job Management

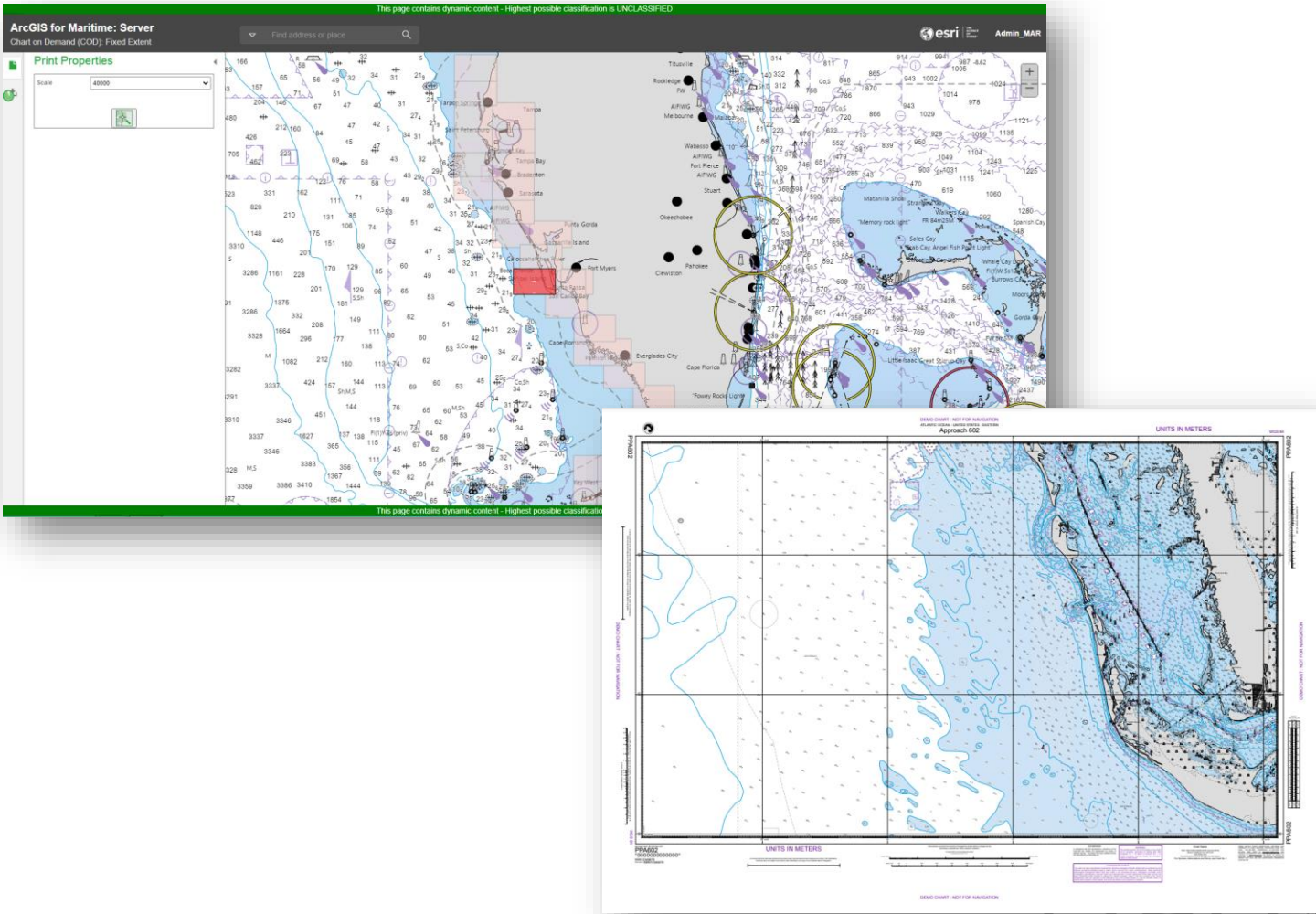


Charts on Demand – On Demand Nautical Chart Production

Capabilities:

- Web-Based Self-Service Chart Production
- S-57 ENC Source Data
- Cartographic Automation
- HACG-like Symbology
- Meets MIL-STD 3060 requirements
- PDF dissemination as Certified Printable ENCs & Nautical Planning Maps

Web Application Data Export

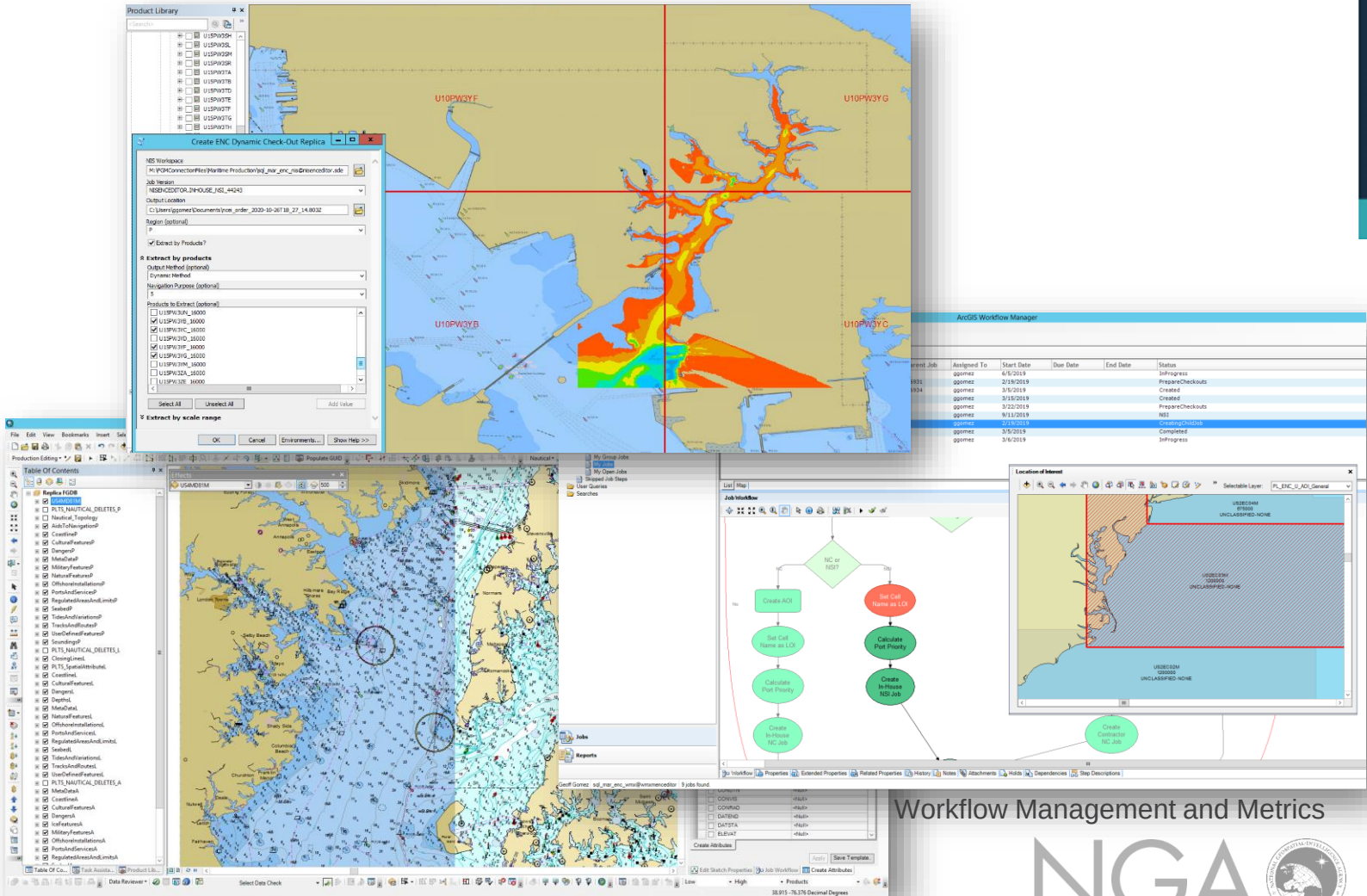


Submarine ENC – Aligning to International Specifications

Capabilities:

- Enterprise Geodatabase
- Virtualized Production Desktop
- Structured Workflows
- Job Management
- S-58 Validation
- S-52 Visualization

Dynamic Checkouts



Production Desktop

Workflow Management and Metrics

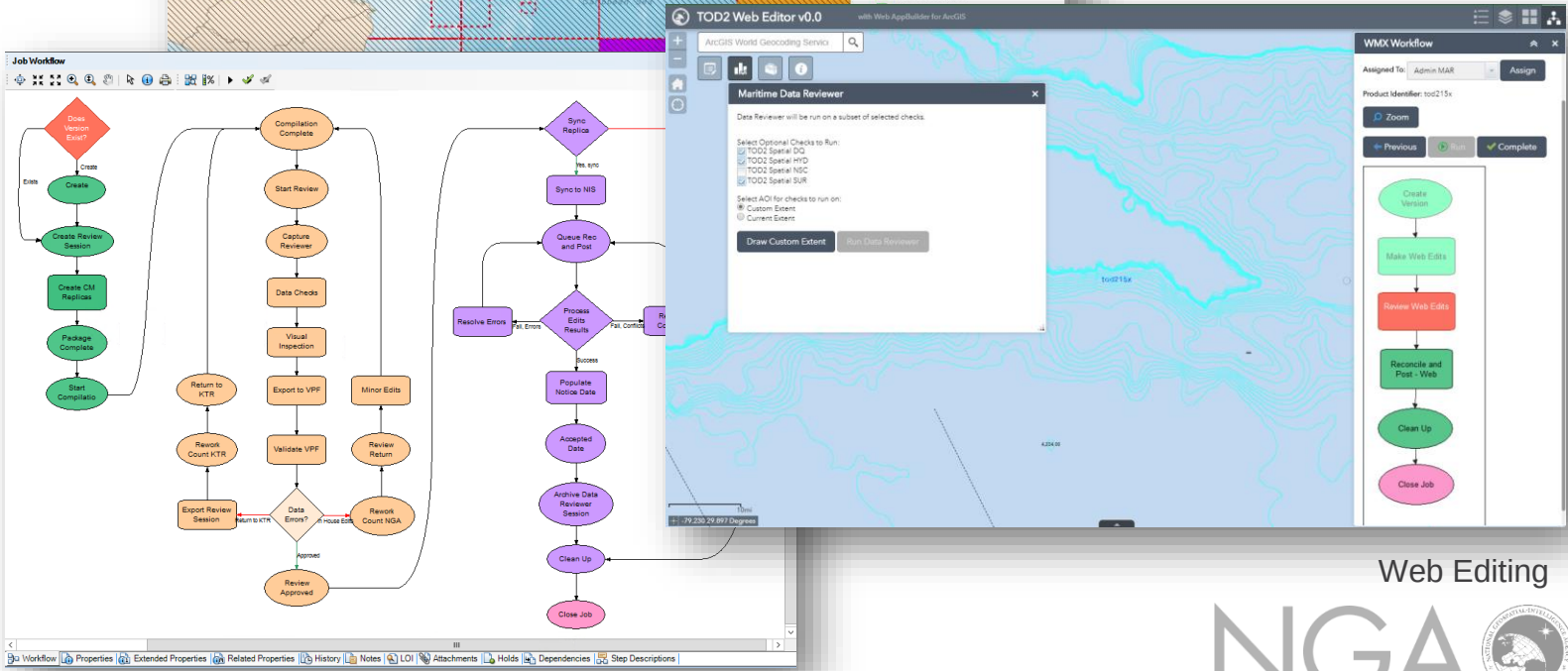
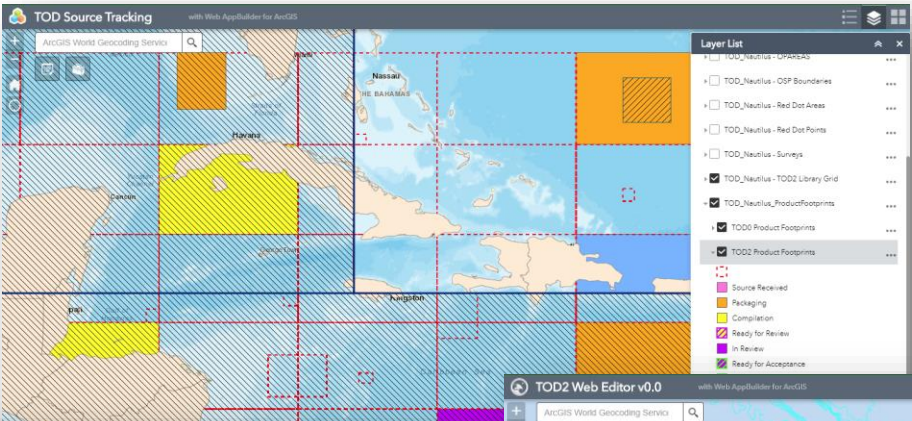


Tactical Ocean Data – Supporting Sub-surface Safety of Navigation

Capabilities:

- Enterprise Geodatabase
- Virtualized Production Desktop
- OPAREA VPF Export
- Bottom Contour VPF Export
- Bathymetric Navigation Planning VPF Export
- Structured Workflows
- Quality Control
- Job Management
- Source Tracking

Source Tracking



Workflow Management and Metrics

Web Editing



Anti-Shipping Activity Messages – Managing and Disseminating Piracy Data

Capabilities:

- Enterprise Geodatabase
- GeoForm Data Entry App
- Quality Control
- Dissemination Services
- Visualization and Spatial Analysis Dashboard

Ops Dashboard



The 'Add an ASAM Event' form has three main sections: 1. Enter Information (with fields for Name, Date, and checkboxes), 2. Select Location (with a world map), and 3. Complete Form (with a Submit button).

GeoForm



Mobile Offshore Drilling Units – Managing and Disseminating Exploration Facilities

Capabilities:

- Enterprise Geodatabase
- Web Editing Application
- Quality Control
- Dissemination Services
- Visualization and Download Web App

Web Editing App

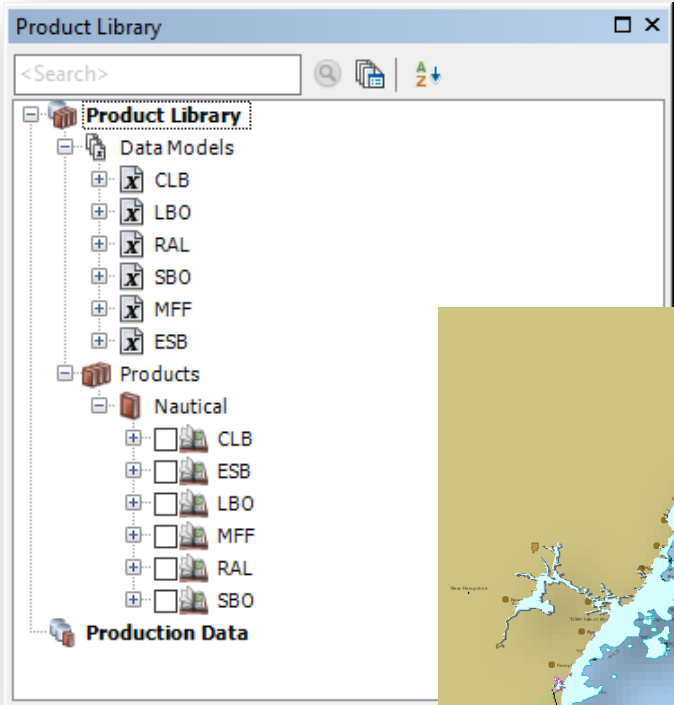


Visualization and Download App

Additional Military Layers – Aligning to International Specifications

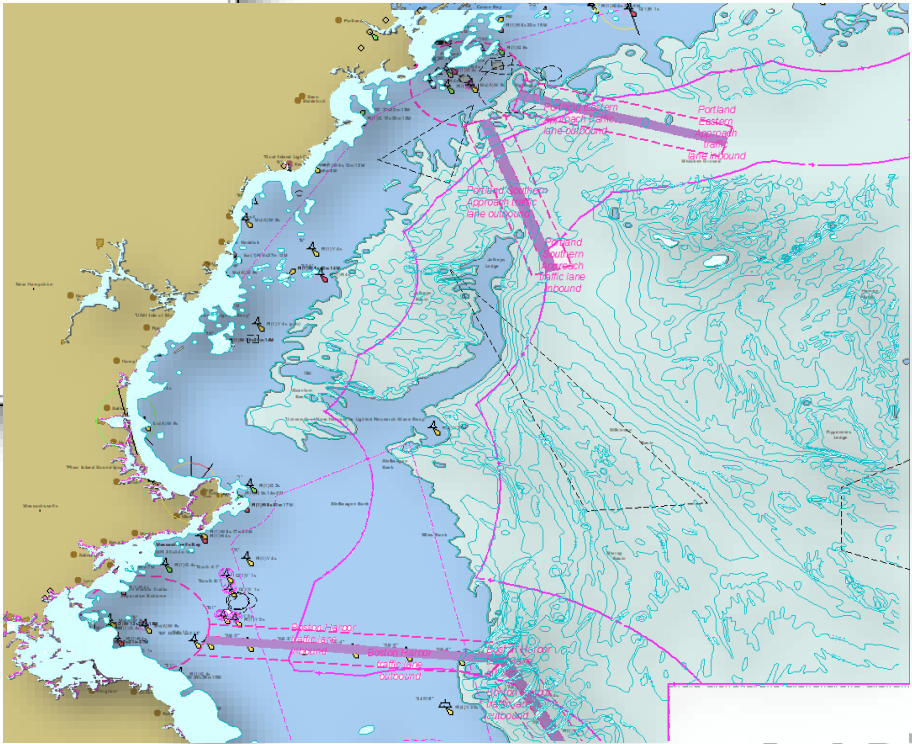
Capabilities:

- File Geodatabases
- Virtualized Production Desktop
- S-52 Visualization



Product Management

CLB



RAL

Radio Navigational Aids – Marine Radio and Electronic Nav Services

Capabilities:

- Enterprise Geodatabase
- Virtualized ArcGIS Pro Production Desktop
- Visualization Web Application
- Dissemination Services and Data Download

Data Extraction

Radio Navigational Aids Data Download

Input

Output

File Format*

File Geodatabase

Download Type*

Map View

☐Download All Layers

☒Select Layer to Download

Layer to Download*

Broadcast Station List

To download a customized group of features, click the Download button on the widget toolbar, located on the widget toolbar in the right section of this application. Next, click the Download button on the widget toolbar, located on the widget toolbar in the right section of this application. Next, click the Download button on the widget toolbar, located on the widget toolbar in the right section of this application.

To download features in a specific area, select an area of interest on the map using one of the area of interest drawing tools. The area of interest will be applied to the download.

If none of these options are used, all features will be included in the download. Click "Start Download" below, to start the download.

Choose Download Extent:

☒Worldwide

☐Area of Interest

File Name*

RadioNavAids_data_download

Start Download

Reference Publication

GMDSS, PIRACY AND U.S. NAVY ASSISTANCE FOR EMERGENCY SITUATIONS

400DG, Peru



MRCC Callao

Contact: MMSI: 007600125, Phone: +0051-1-4200177, Fax: +0051-1-4299798, E-mail: comera.callao@dcaps.mil.pe

DSC Station Callao

MMSI: 007600125

Station Type: VHF (Main) range 50nm

MF (Main) range 200nm

HF on 8 MHz

Location: 12-03S 077-09W

Monitor Times: 24-7

Visualization Web App

21

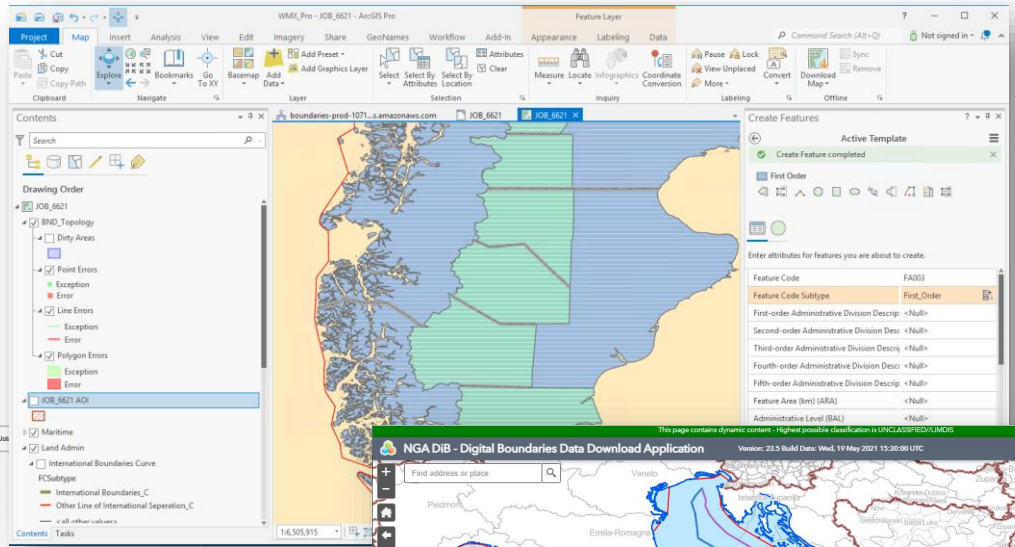
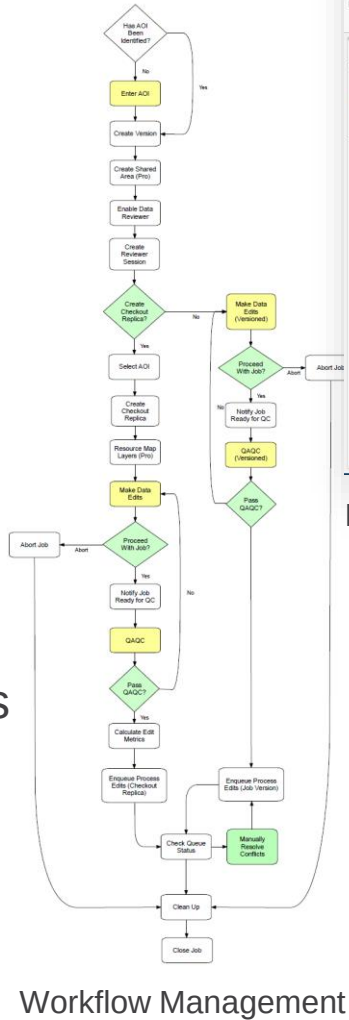
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Office of Geography

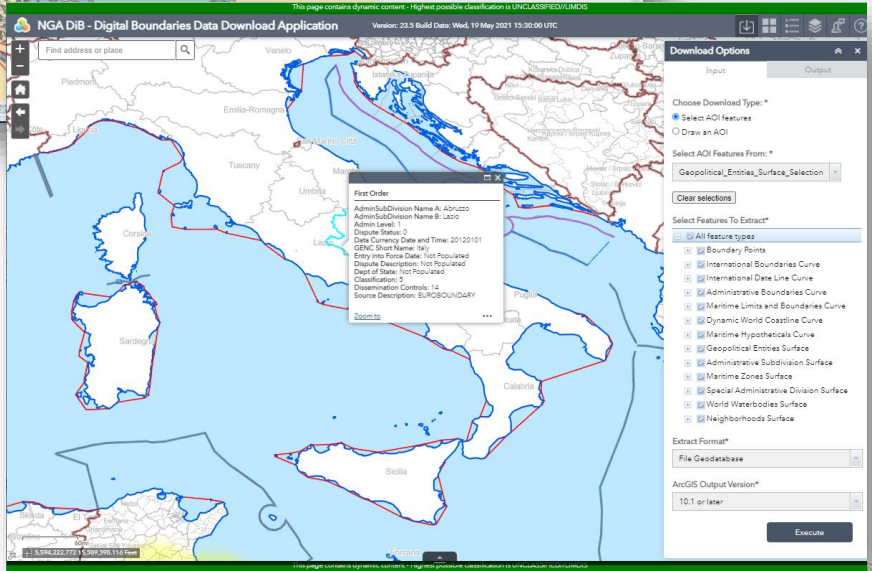
International Boundaries and Limits – Integrated Land and Maritime Production

Capabilities:

- Enterprise Geodatabase
- Virtualized Production Desktop
- Structured Workflows & Tasks
- Job Management
- Quality Control
- *Operations Dashboard*
- *DiB Web Portal (Sites Page)*
- Dissemination Web Map Services
- Data Visualization & Download Web App



Desktop Production

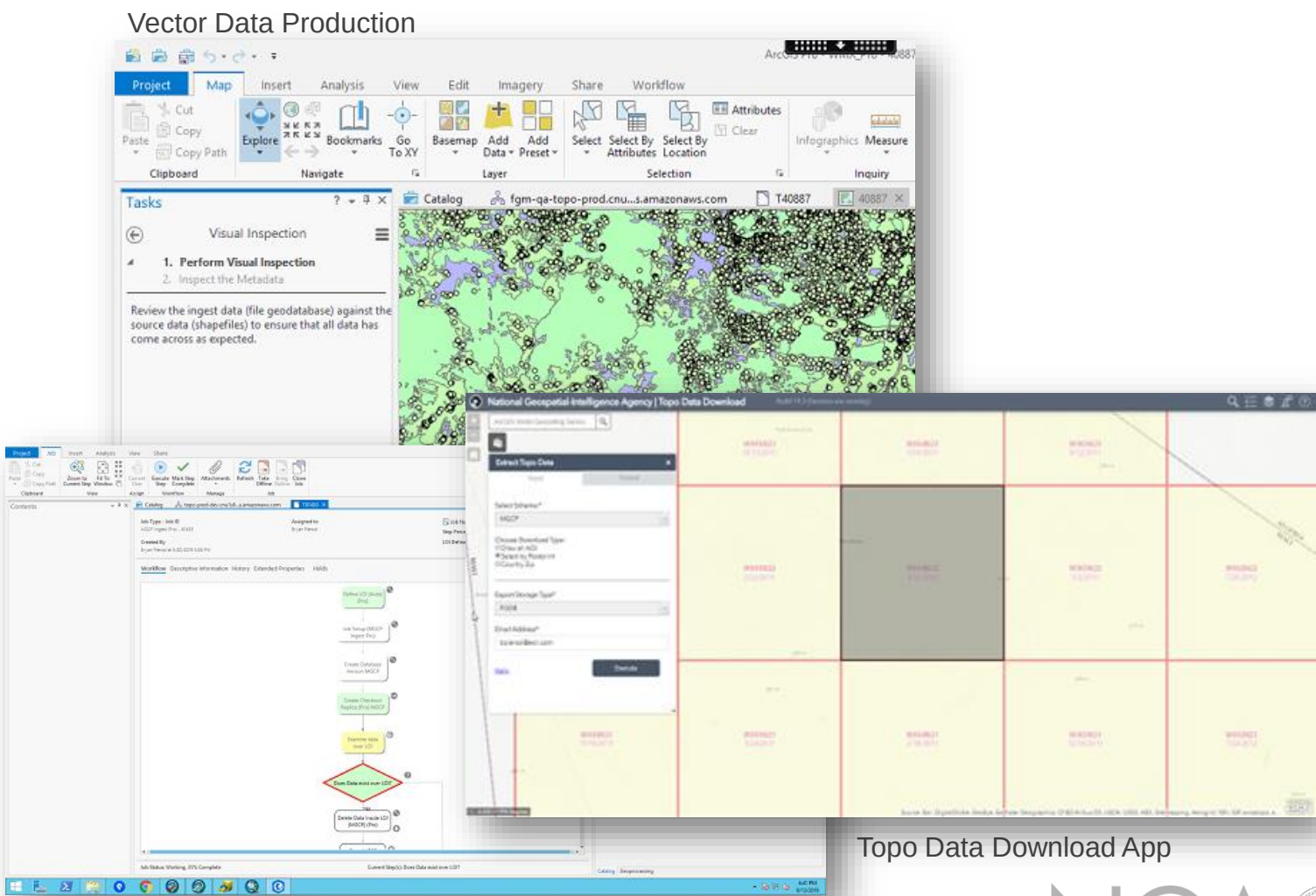


Data Visualization & Download Web App

Topography – Vector – Enterprise Topographic Data Management

Capabilities:

- Enterprise Geodatabases
 - TDS, MGCP, RPM, RPMi, VISNAV, VISNAVHD, HRVD, MUVD,
- Virtualized Production Desktop
- Structured Workflows
- Job Management
- Quality Control
- Dissemination Services
- Topo Data Download



Workflow and Job Management

Topo Data Download App

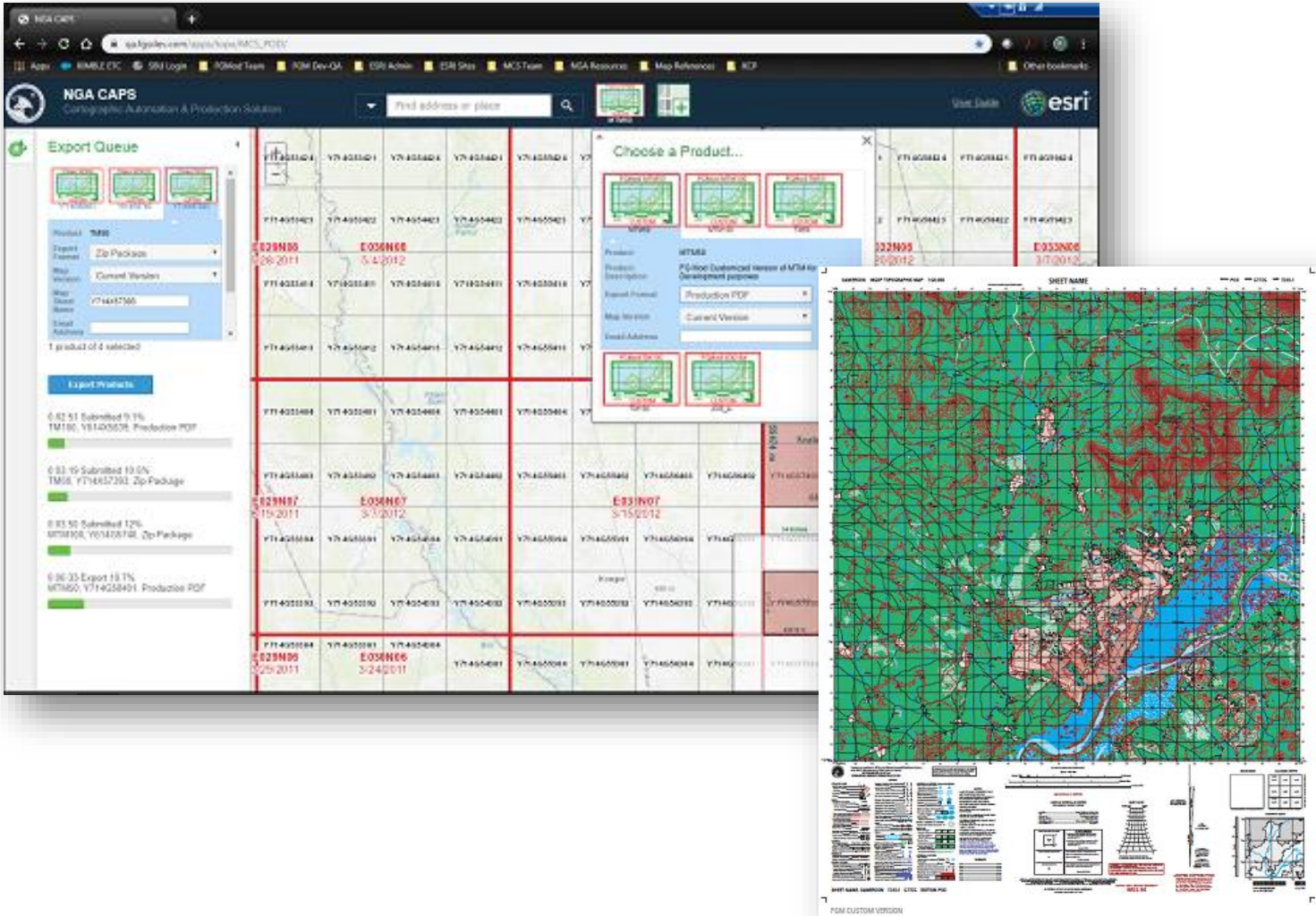


Topography – Cartography – Automated Cartographic Production

Capabilities:

- Enterprise Data Integration
- Cartographic Automation
- Web-based Cartographic Production
- Virtualized Production Desktop for finishing

Web-based Cartographic Production



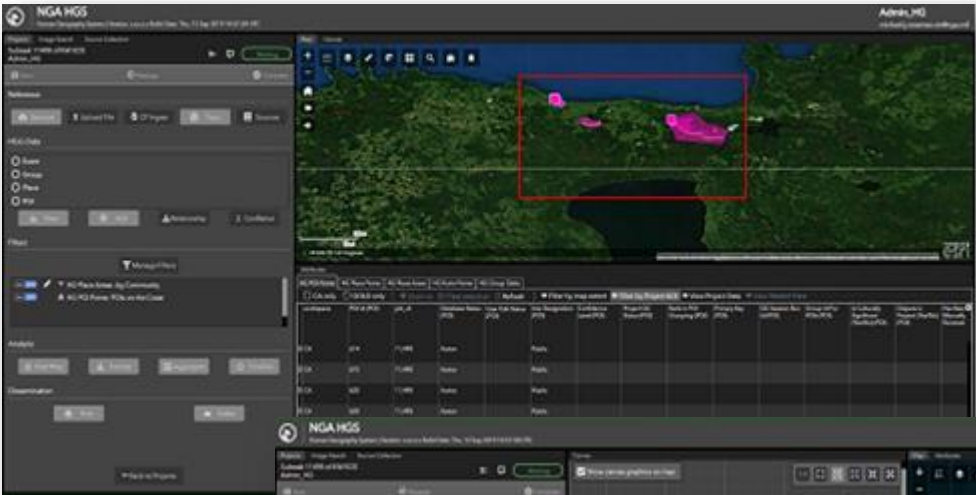
NGA Product Specifications

Human Geography – Integrated and Centralized HG Production

Capabilities:

- Centralized Data Management
- Web Based Production App
- Workflow & Job Management
- Imagery Search and Discovery
- Relationship Visualization
- Quality Control
- Dissemination Services

Web Based Production



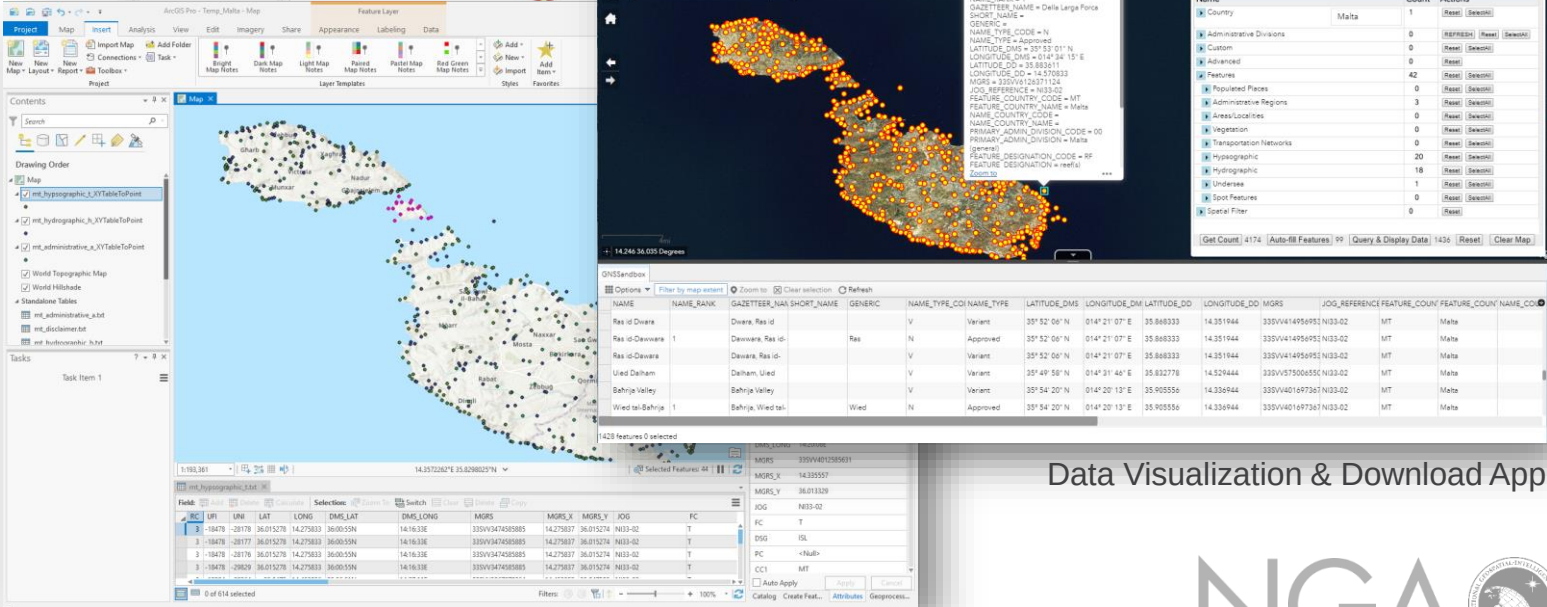
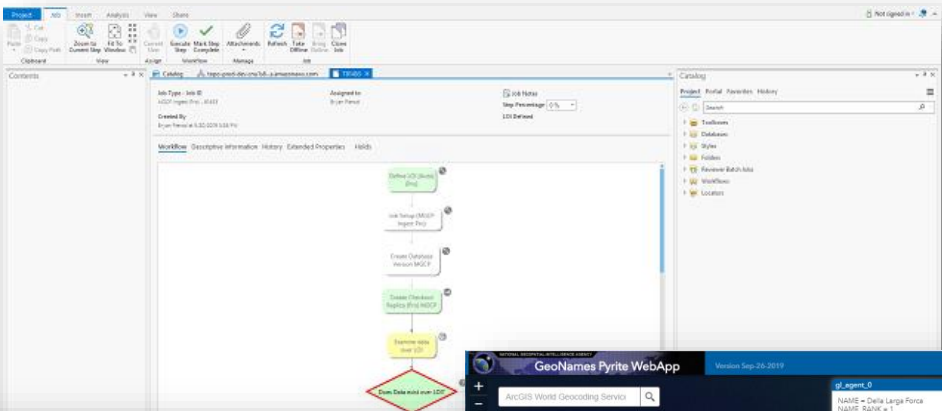
Relationship Canvas

Geographic Names – Official U.S. Foreign Geographic Names Data Management

Capabilities:

- Enterprise Geodatabase
- Virtualized Production Desktop
- Structured Workflows, Job Management & Quality Control
- Production Status and Data Quality Dashboards for visualization and planning
- Dissemination Services (NGA and public-facing)
- Names Data Visualization and Download Web App
- GNS Website internationalized in at least 6 languages

Workflow Management



Data Visualization & Download App

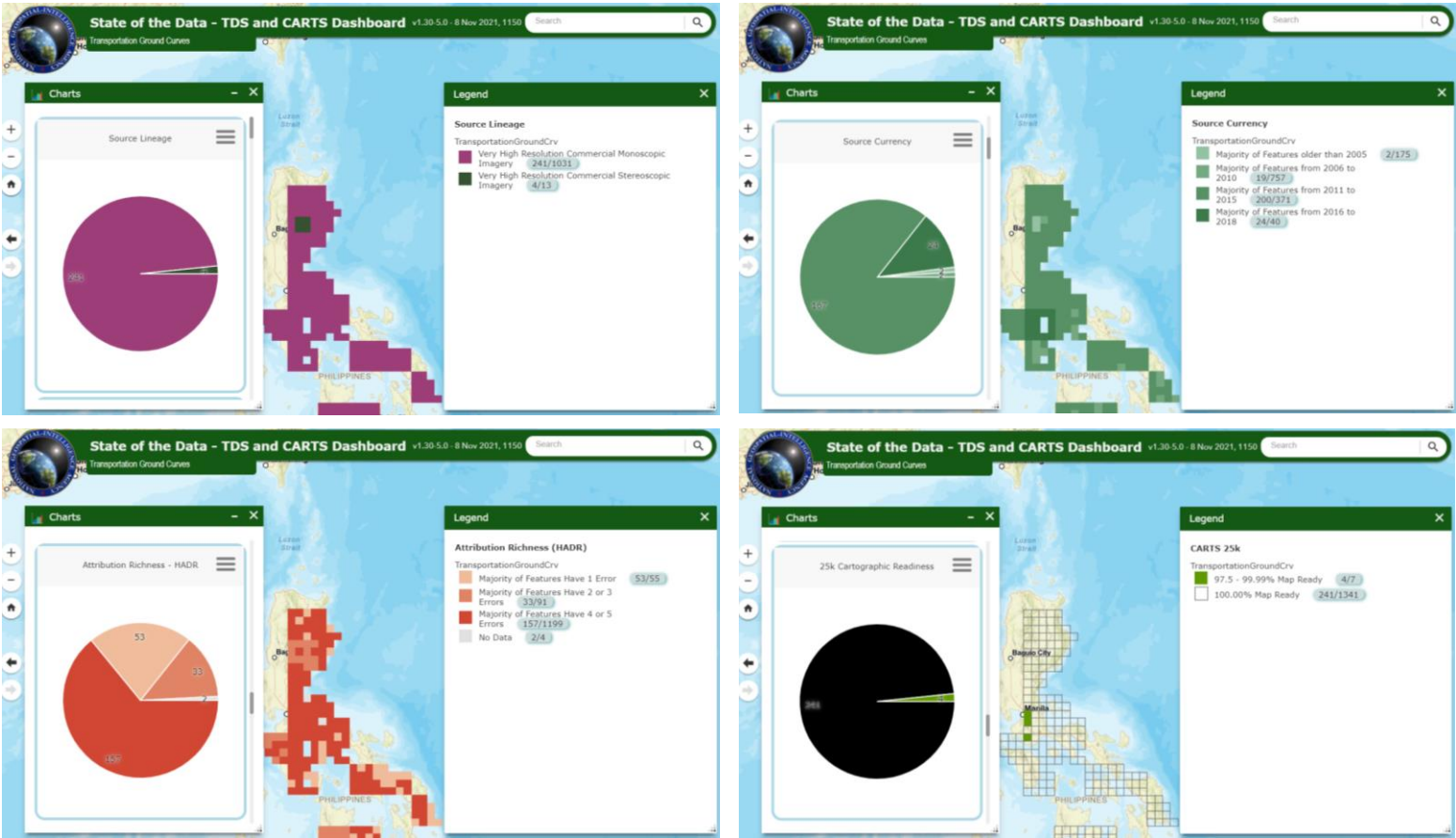
Desktop Production



State of the Data- Understanding the quality and usability of topographic data

Capabilities:

- Statistically quantifying the quality of TDS data
- Determining the cartographic readiness of TDS data
- Leveraging big data technology to process ~ 1 billion topo features in less than a day
- Content assessments delivered as web-based services
- All consumed through a single web-based dashboard



Enabling pinpoint content management by developing quantitative metrics to explain data quality and cartographic readiness



GeoAI– Automated Feature Extraction using Machine Learning

Capabilities:

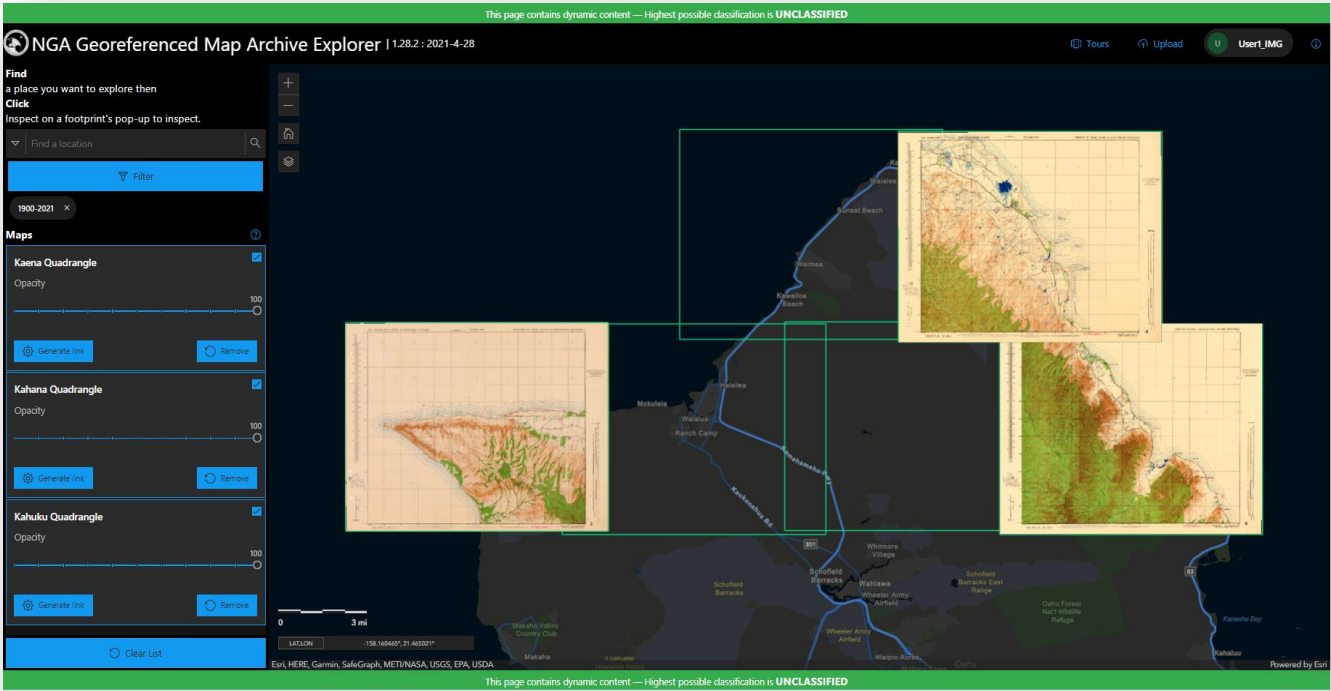
- Explore GeoAI and deep learning to automate feature extraction
- Utilize Esri’s tools and pretrained deep learning models as starting point
- Automate extraction of building & road geometries and attribution
- Leverage NGA imagery services and topo data for model training
- Identify differences with existing data and conflate into authoritative holdings to create map ready data
- Utilize Cloud computing resources in NGA environments
- Focus on capabilities that can be operationalized



Georeferenced Map Archive – Crowdsourced Georeferenced Maps

Capabilities:

- Web Mapping Application
- Upload & Store Georeferenced Maps
- Metadata Catalog for Advanced Search
- View Thumbnails
- Download Georeferenced Maps
- ArcGIS Pro Add-In
- Deployed to UC2S



Georeferenced Map Archive Explorer

Office of Geomatics

Precise Imagery – Dynamic Image Services for the Enterprise

Capabilities:

- Large, Diverse Imagery Catalog
- Ortho on-the-fly Image Services
- Elevation Services
- Read from Cloud Storage
- Catalog Search Widget
- Exploitation in Desktop and Web Applications

Desktop Exploitation

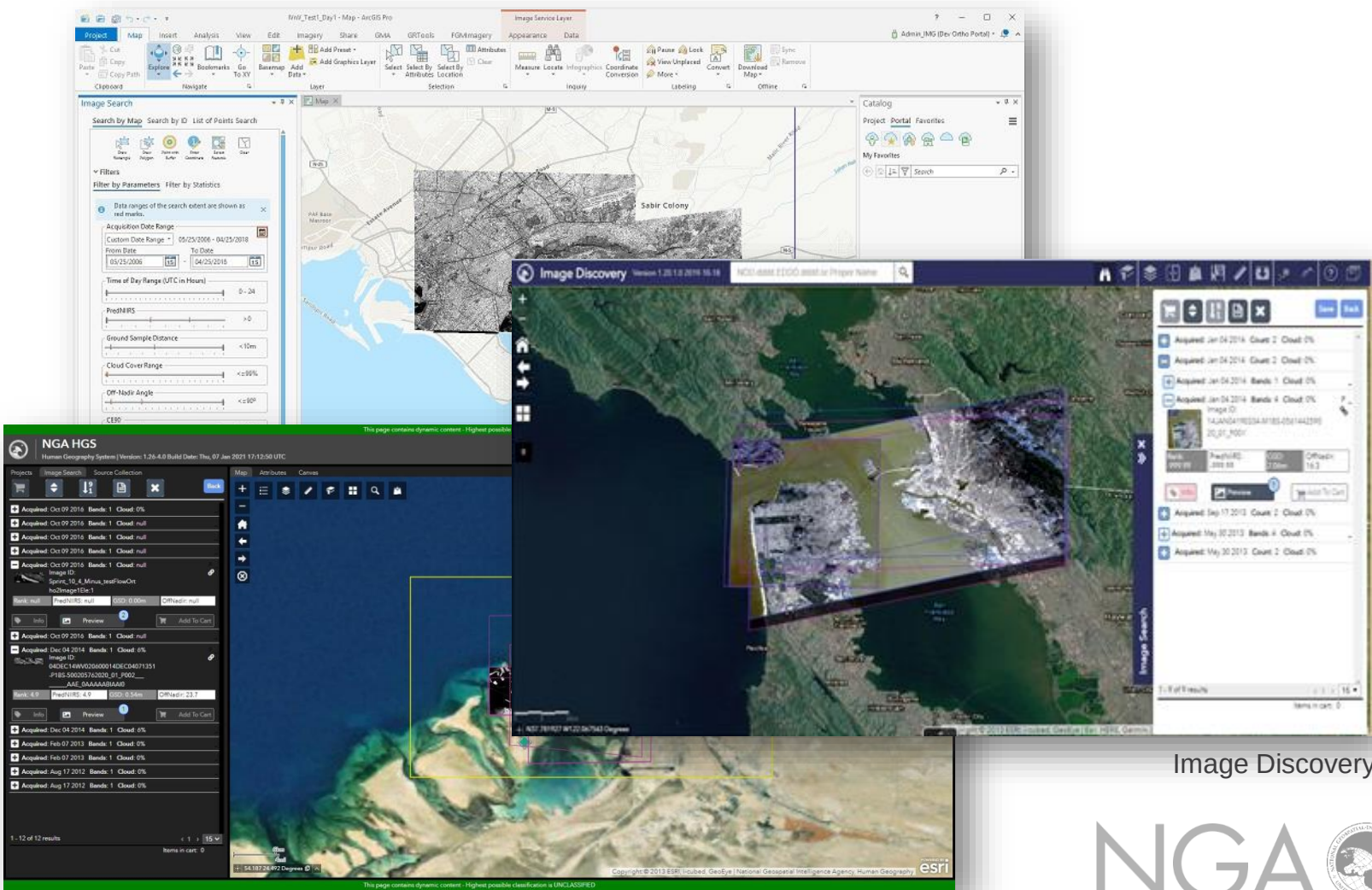


Image Discovery



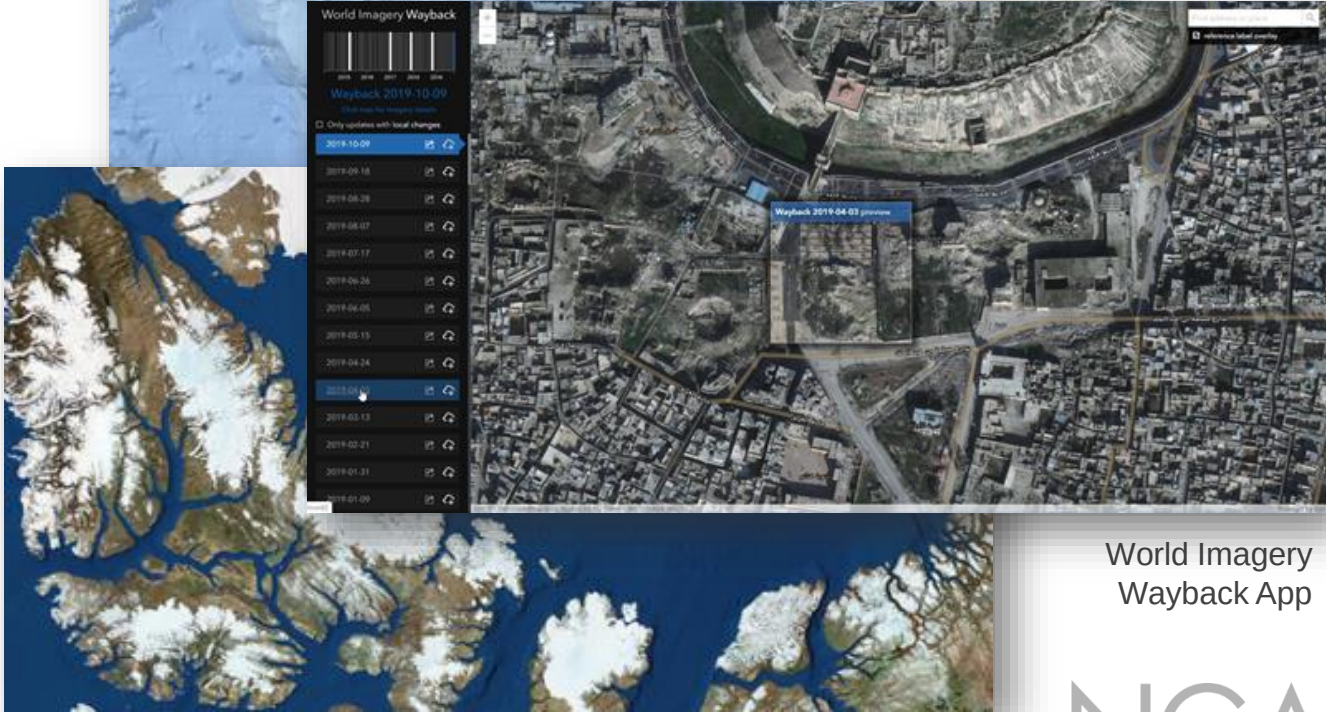
Web-Based Exploitation

World Imagery Wayback – Data Appliance for ArcGIS

Capabilities:

- Commercial Imagery WMTS Tile Layers
- Over 90 versions spanning 5+ Years
- Deployed to C2S
- Wayback JavaScript Web Application
- Living Atlas of the World Content

Oceans Basemap



World Imagery
Wayback App

Arctic Imagery Basemap

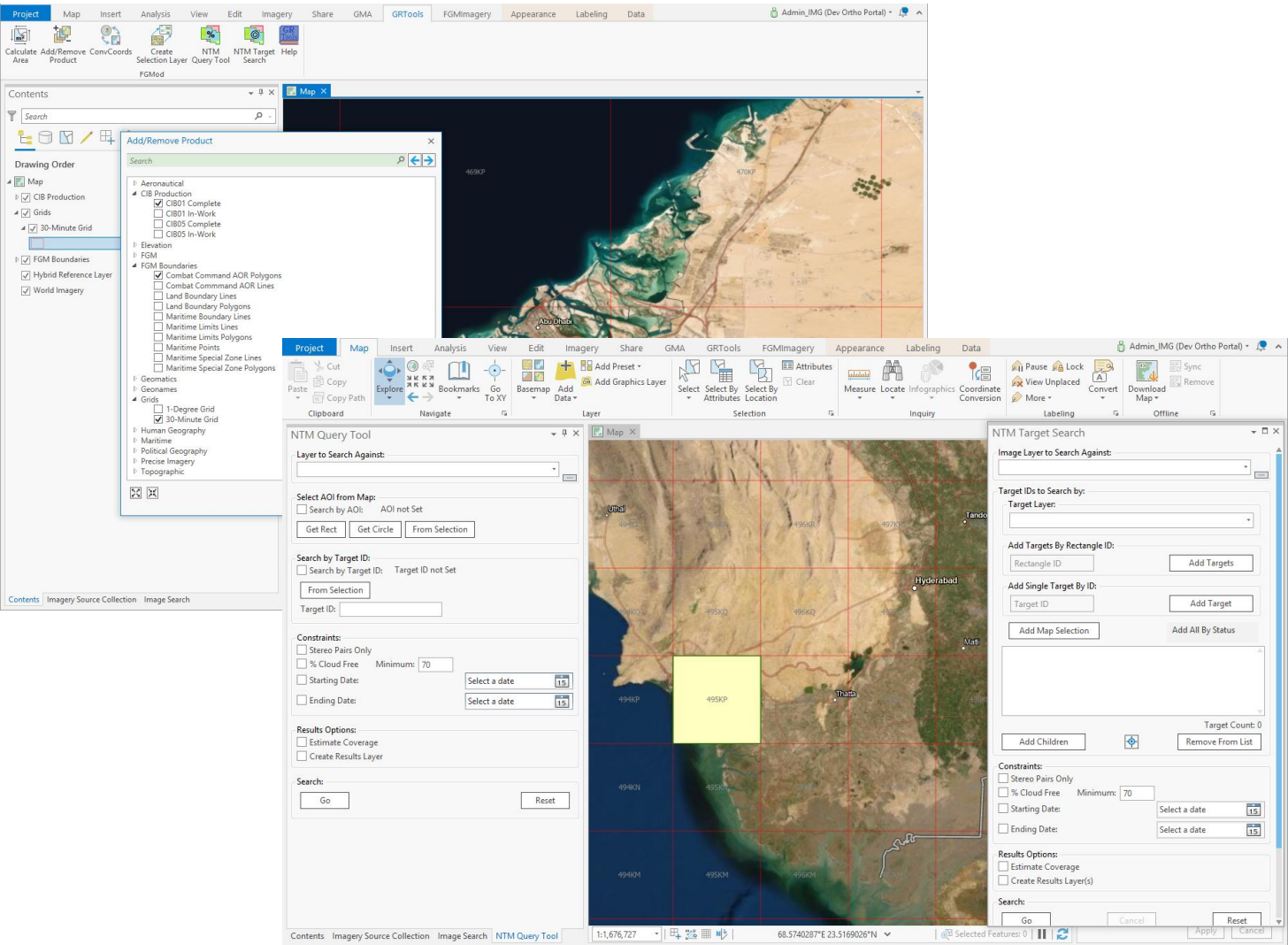


FGM GRTools – Modernized GEOINT Research Tools

Capabilities:

- Extension to ArcGIS Pro Desktop
- Textual and Spatial Search of:
- Commercial and NTM Imagery
- NTM Target Layers
- Elevation
- Boundaries
- CIB and DPPDB Production
- Status Layers
- Custom data layers provided by users
- Export of search results to JSON, text files, or shapefiles for use in other systems
- Analysis of Commercial and NTM imagery collections for CIB and DPPDB production

Desktop Image Search of Foundation GEOINT Layers



NTM Imagery and Target Layer Search Tools

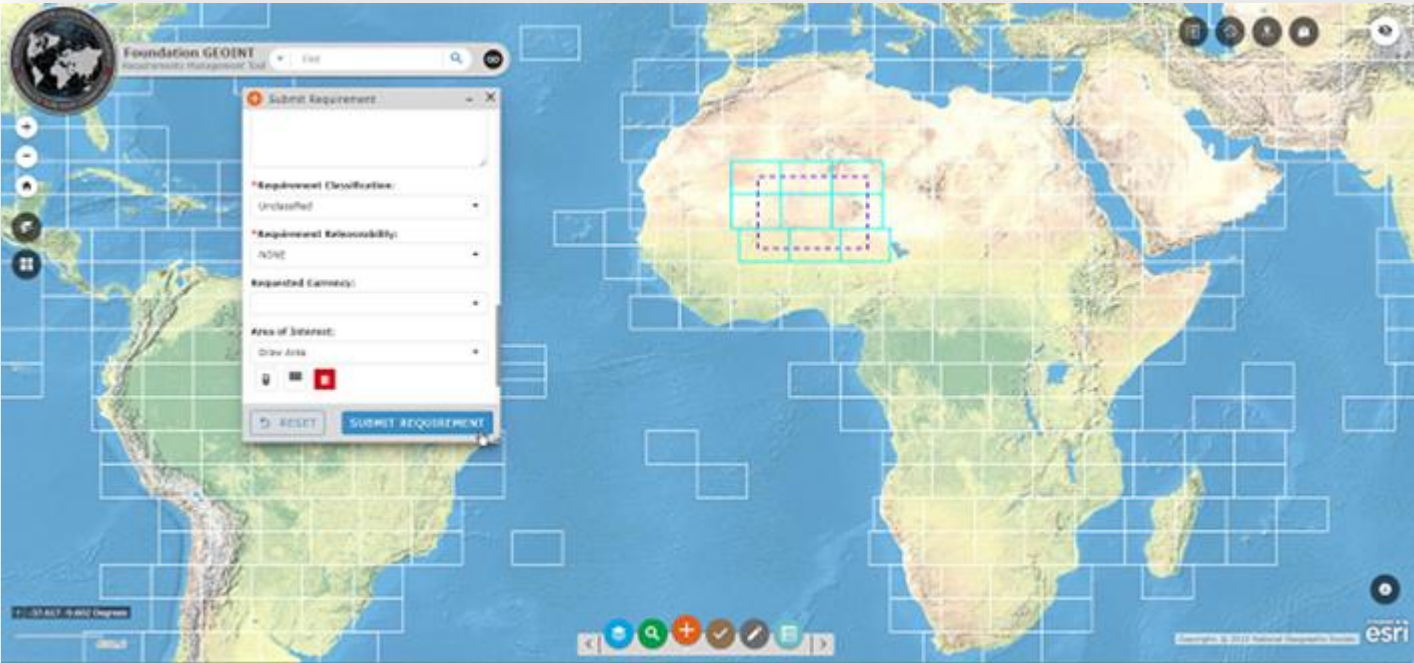
NOX Office

Requirements Management – Collecting and Managing Production Requirements

Capabilities:

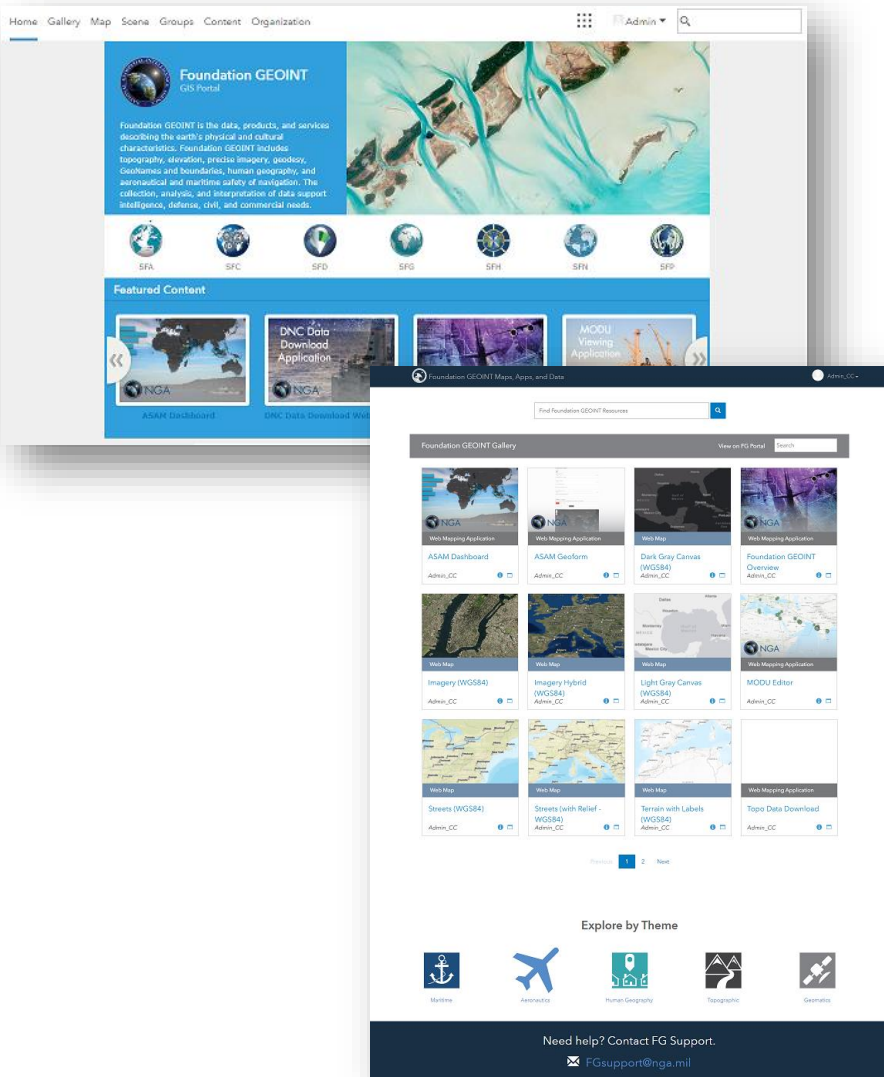
- Web-based Requirements Collection
- Virtualized Desktop for Requirements Management
- Integration with Production Office Workflows
- Requirements Tracking and Reporting

Requirements Collection

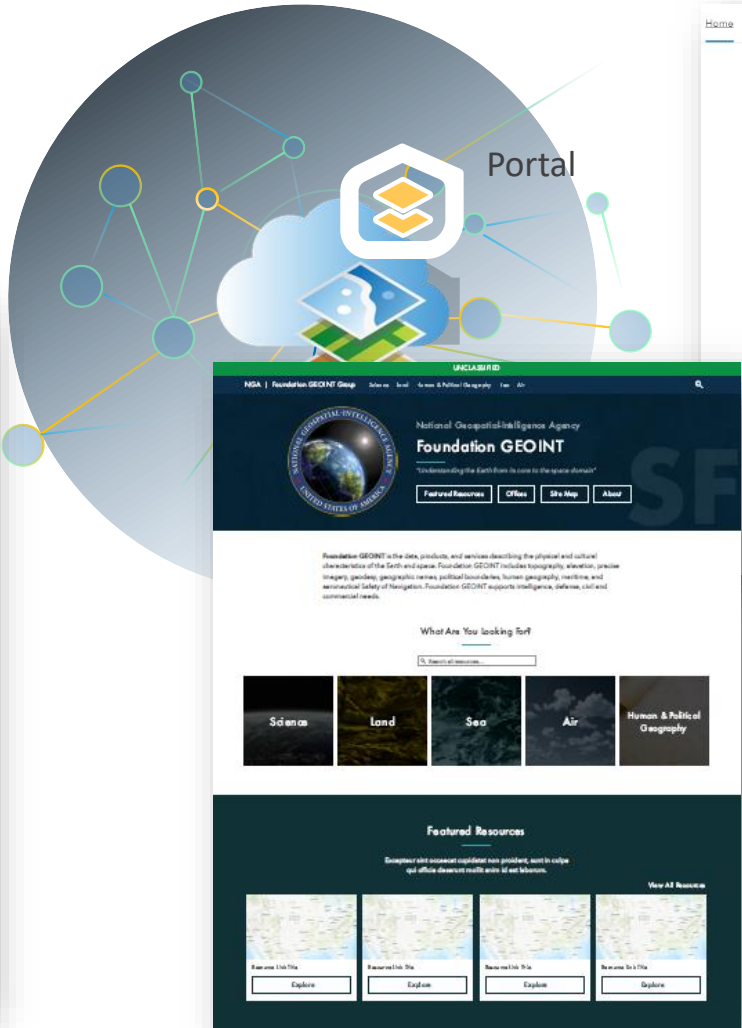


Data Discoverability- Supporting NGA's System of Engagement

Foundation GEOINT Portal

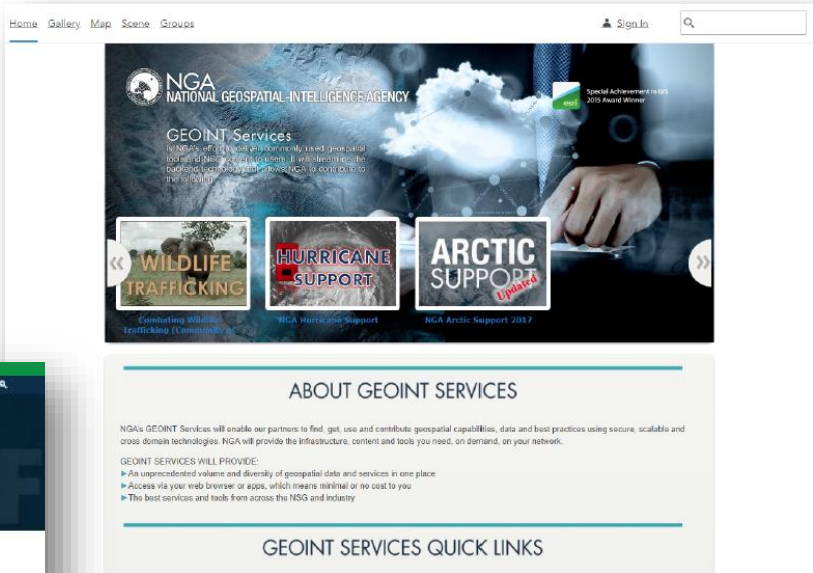


Foundation GEOINT Gallery



Foundation GEOINT Sites

IC GIS Portal



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